

Academic Curriculum Vitae

Timothy Charles Andrews

September 16, 2026

Place of Birth: Croydon, England

Nationality: Dual citizen of New Zealand and the United Kingdom

New Zealand Permanent Residence: 28 Plympton St, New Plymouth, 4310

United States Current Residence: 1105 Nielsen Court, Apt 6, Ann Arbor, Michigan, 48105

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Research Interests

- Numerical modelling of geophysical fluid equations
- Development and testing of dynamical cores for Global Circulation Models (GCMs)
- Diffusion mechanisms in dynamical cores
- Analysis and comparison of timestepping methods
- The phase-averaged timestepping method for oscillatory PDEs

Academic Positions

Postdoctoral Research Fellow - Department of Climate and Space Science, University of Michigan, United States of America (October 2024 – October 2027)

Education

Doctor of Philosophy (PhD) in Mathematics (Geophysical Fluid Dynamics) — University of Exeter, United Kingdom (September 2020 – September 2024)

Bachelor of Engineering with First Class Honours, specialising in Engineering Science — University of Auckland, New Zealand (2016 – 2019)

Selected Presentations

- Poster presentation at EGU 2026: **Andrews, T.**, Jablonowski, C., Hughes, O., and Bendall, T.: New mountain-generated mesoscale flow test cases from DCMIP2025, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-2009, <https://doi.org/10.5194/egusphere-egu26-2009>, 2026.
- Poster presentation at EGU 2025: **Andrews, T.**, and Jablonowski, C.: New mountain-generated mesoscale flow test cases for dynamical cores. AGU Fall Meeting 2025, New Orleans, USA.
- Oral presentations at the 2025 and 2026 Community Earth System Model (CESM) workshops (Boulder, United States)
- Oral presentation at the 2025 ‘PDEs on the Sphere’ conference (São Paulo, Brazil).
- Oral presentation at the Parallel-in-Time (PinT) 2023 workshop (Hamburg, Germany)
- Oral presentation at the 2023 ‘PDEs on the Sphere’ conference (Grenoble, France)

- **Andrews, T.**, Wingate, B., and Shipton, J.: Quantifying the accuracy of large time-steps in highly oscillatory systems, EGU General Assembly 2023, Vienna, Austria, 23–28 Apr 2023, EGU23-14086, <https://doi.org/10.5194/egusphere-egu23-14086>, 2023.
- Visiting seminar at the National Institute of Water and Atmospheric Research (NIWA), 2023 (Wellington, New Zealand)

Scientific

Peer-Reviewed Articles

- **Andrews, T. C.**, and C. Jablonowski (2026): Stability Analyses of Divergence and Vorticity Damping on Gnomonic Cubed-Sphere Grids. *Mon. Wea. Rev.*, 154, 1317–1336, <https://doi.org/10.1175/MWR-D-25-0192.1>.
- **Andrews, T. C.**, Wingate, B.A. (2026): A numerical study of mean corrected phase-averaging in geophysical flow equations. *Theor. Comput. Fluid Dyn.* 40, 28. <https://doi.org/10.1007/s00162-026-00797-x>

Manuscripts in review

- **Andrews T. C.**, Jablonowski C., and Hughes O. K., and Bendall T. M. (2026), ‘A mountain-generated mesoscale test case from DCMIP-2025: Gap flow and vortex shedding variants’. Submitted to Geophysical Model Development, available online at <https://egusphere.copernicus.org/preprints/2026/egusphere-2026-2293>.
- **Andrews, T. C.**, Bendall, T. M. (2026) ‘A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements.’ Submitted to the Journal of Computational Physics and available as an arXiv preprint at <https://arxiv.org/abs/2603.19075>.
- **Andrews T. C.**, Duru K., Lee D., (2026) ‘A perfectly matched layer for damping vertically propagating waves in the nonlinear, compressible Boussinesq equations’. Submitted to the Journal of Computational Physics and available as an ArXiv preprint at <https://arxiv.org/abs/2606.28303>.
- Pletzer, T., Pletzer, A., **Andrews, T. C.**, Lin, D., and Katurji, M.: Mimetic interpolation for finite-time Lyapunov exponent computation near irregular domain boundaries, EGU sphere [preprint], <https://egusphere.copernicus.org/preprints/2026/egusphere-2026-3377>, 2026.

Preprints

- **Andrews, T. C.**, Shipton, J., Wingate, B. (2023), ‘The effect of linear dispersive errors on nonlinear timestepping accuracy in the f-plane rotating shallow water equations’. Available as an arXiv preprint at <https://arxiv.org/abs/2305.06685>.

In Preparation

- **Andrews, T. C.**, Jablonowski C., (2026), ‘A study of momentum equation diffusion in the CAM-FV3 dynamical core using baroclinic wave simulations’. In preparation for submission to Geophysical Model Development.

PhD Thesis

PhD Thesis from the University of Exeter: ‘Using nonlinear interactions to quantify and improve timestepping accuracy in the Rotating Shallow Water Equations’ (2024). Available at: <https://ore.exeter.ac.uk/articles/thesis/29811980?file=56858228>.

Other Research Experience and Skills

- A code developer for the Gusto, a compatible finite element library for solving the geophysical fluid equations (<https://www.firedrakeproject.org/gusto>).
- Connections with the UK Met Office, including code contributions to the trunk of LFRic’s shallow water mini app.

- Collaborations with Alex Pletzer at NIWA (New Zealand) for mimetic vector interpolation for LFRic, including comparing different Jacobian approximations.
- I can use the programming languages of Python, Fortran, MATLAB, R, and Visual Basic.
- Summer Internship research project at the Methanex New Zealand plant in Taranaki. This focused on using extreme value statistics to extrapolate the states of the plant's processing devices, given a small testing sample (2018-2019).
- Summer Research Scholarship project in the Precision Acoustic Group, within the University of Auckland Physics Department. I investigated using reflected acoustic measurements to estimate soil moisture properties (2017-2018).

Academic Service

- Volunteer peer reviewer for AMS Journals (2025).
- Instructor at the DCMIP-2025 summer school (<https://sites.google.com/umich.edu/dcmip-2025>) (2025).
- Judge for the Michigan Geophysical Union annual symposium (2025).
- Coordinator for scientific committee elections for the Parallel-in-time (PinT) community (2023, 2025).

Academic Achievements

During my bachelor's degree in Engineering Science:

2019

First in Course Awards for:

ENGSCI 711: Advanced Mathematical Modelling

ENGSCI 740: Advanced Mechanics in Research and Technology

ELECTENG 733: Signal Processing

2018

First in Course Awards for:

ENGSCI 314: Mathematical Modelling 3

ENGSCI 343: Mathematical and Computational Modelling in Mechanics

ENGSCI 344: Modelling and Simulation in Computational Mechanics

ENGSCI 391: Optimisation in Operations Research

MECHENG 325: Dynamics of Fluids and Structures

2017

First in Course Awards for:

ENGSCI 233: Computational Techniques and Computer Systems

ENGSCI 255: Modelling in Operations Research

2016 – 2018

Named on the University of Auckland Dean's Honours List for Top Engineering Students of the Year

Last year of High School:

2015

Dux of New Plymouth Boys' High School (NPBHS)

NZQA Scholarships attained in Physics, Statistics, and Economics

Awarded an IPENZ Foundation Scholarship to study Engineering

Teaching

- Guest lecturer for CLIMATE401: Geophysical FLuid Dynamics, University of Michigan (2025)
- Teaching assistant for NSC1002 Mathematics and Computing, University of Exeter (2022).
- High School maths and physics tutor for Taranaki Tutors, New Plymouth, New Zealand (2020–2022).

Other Achievements

- Ran the Wanganui marathon (42km) in under three and a half hours (2019)
- Have completed five half marathon runs
- Completed the ‘Lake Taupo Cycle Challenge’ (Approx. 160km) (2018)
- Member of the Auckland University Division 1 Men’s hockey team (2016-2017)
- Member of the 1st XI NPBHS Hockey team (2013-2015)
- Member of the NPBHS Senior Debating Team (2015)

Hobbies

I am passionate about outdoor activities such as hiking (we call it tramping in New Zealand!) and running. I love planning and executing multiple-day adventures in the forest parks of New Zealand. I am also aiming to complete the New Zealand Coast to Coast challenge sometime. In my spare time, when I’m not enjoying nature, I relax by cooking and reading epic fantasy books.

Academic Referees

Professor Beth Wingate - Department of Mathematics and Statistics, University of Exeter (Emeritus)
PhD supervisor
b.wingate@exeter.ac.uk

Professor Christiane Jablonowski - Department of Climate and Space Sciences and Engineering, University of Michigan
Postdoctoral research supervisor
cjablono@umich.edu